

Central Heating Systems: Circulating Pumps and Open-Vented Systems

Introduction

In a central heating system, hot water must circulate continuously between the boiler and the heating appliances. In an open-vented system, the position of the open vent and cold feed is important for safe and effective operation. Once these have been considered, attention can be given to another important component: the circulating pump.

The circulating pump moves water around the heating circuit. In many diagrams, the pump is shown on the flow pipe leaving the boiler. However, in practical installations, the pump may also be installed on the return pipework. The important point is that the pump must circulate water in the correct direction around the system.

The Function of the Circulating Pump

The main purpose of the circulating pump is to create a pressure difference that causes water to flow through the heating system. The boiler heats the water, and the pump circulates it through the pipework to the radiators or other heating components.

The basic circulation can be described as:

Boiler → Flow Pipe → Radiators → Return Pipe → Boiler

As hot water passes through the radiators, it releases heat into the rooms. The water then becomes cooler and returns to the boiler, where it is heated again. This process continues while there is a demand for heating.

Pump on the Flow

A common arrangement places the pump on the flow pipe immediately after the boiler. In this arrangement, the heated water leaves the boiler and passes through the pump before travelling around the heating circuit.

The pump helps move the hot water away from the boiler and through the radiators. This arrangement is often shown in educational diagrams because it makes the direction of circulation easy to understand.

However, it is important not to assume that every heating system must have the pump in this position.

Pump on the Return

The circulating pump can also be located on the return pipework. In this arrangement, water leaves the boiler, travels around the heating circuit, and then passes through the pump before returning to the boiler.

The pump still performs the same basic function: maintaining circulation around the complete system.

Therefore, whether the pump is installed on the flow or return, the key consideration is that it must be correctly positioned and oriented according to the system design.

Open Vent and Cold Feed

In an open-vented system, the open vent and cold feed are important for expansion and safety. Water expands when heated, so the system needs an arrangement that can accommodate this expansion.

The open vent connects the heating system to the atmosphere, while the feed-and-expansion cistern provides a means of supplying water and accommodating expansion.

The pump's position must be considered carefully in relation to these components. Poor system design can cause a condition known as **pumping over**, where water is forced up the open vent and into the cistern. This can lead to water loss, aeration, corrosion, and other problems.

Conclusion

The circulating pump is essential for maintaining water movement in a pumped central heating system. The boiler provides the heat, while the pump circulates the heated water around the circuit.

Although diagrams often show the pump on the flow pipe, it may also be installed on the return pipework in practical systems. What matters most is that the pump operates in the correct direction and works properly with the boiler, pipework, open vent, and cold feed.

Understanding these relationships is essential for designing, installing, and maintaining an efficient and reliable central heating system.